BASE CHANGE FOR INTERNAL HOM FROM PERFECT OR PSEUDO-COHERENT OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof states the precise base change assertion, reduces via the single-complex base change condition and its inheritance, and refers the stalk verification to the argument of the tensor base change lemma in the same chapter

Editorial note. The label lemma-base-change-RHom names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the single-complex base change lemma, its condition, and the inheritance of the condition (lemma-single-complex-base-change, lemma-single-complex-base-change-condition, lemma-single-complex-base-change-condition-inherited of the same chapter)
- the stalk verification that for a complex G flat over Y the canonical map L(g')^*G -> (g')^*G satisfies the base change condition, as established in the proof of lemma-base-change-tensor of the same chapter

Source reference. Original reference 0A1D: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5625–5625. Prerequisite pointer invoked at proof line(s) 5720; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DJ9: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5488–5488. Prerequisite pointer invoked at proof line(s) 5712; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DJ8: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5365–5365. Prerequisite pointer invoked at proof line(s) 5719; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DJA: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5567–5567. Prerequisite pointer invoked at proof line(s) 5713; it names a standard fact of the same chapter that the counted argument

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism of algebraic spaces over S. Let E be an object of $D(\mathcal{O}_X)$. Let $\mathcal{G}^\bullet$ be a complex of quasi-coherent $\mathcal{O}_X$-modules. If*

(1) *E is perfect, $\mathcal{G}^\bullet$ is a bounded above, and $\mathcal{G}^n$ is flat over Y, or*
(2) *E is pseudo-coherent, $\mathcal{G}^\bullet$ is bounded, and $\mathcal{G}^n$ is flat over Y,*

then formation of

$$Rf_*R\mathcal{H}om(E, \mathcal{G}^\bullet)$$

commutes with arbitrary base change (see proof for precise statement).

Proof. The statement means the following. Let $g : Y' \to Y$ be a morphism of algebraic spaces and consider the base change diagram

$$\begin{array}{ccc} X' & \xrightarrow{h} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

in other words $X' = Y' \times_Y X$. The lemma asserts that

$$Lg^*Rf_*R\mathcal{H}om(E, \mathcal{G}^\bullet) \longrightarrow R(f')_*R\mathcal{H}om(L(g')^*E, (g')^*\mathcal{G}^\bullet)$$

is an isomorphism. Observe that on the right hand side we do **not** use the derived pullback on $\mathcal{G}^\bullet$. To prove this, we apply Lemmas 0DJ9 and 0DJA to see that it suffices to prove the canonical map

$$L(g')^*\mathcal{G}^\bullet \to (g')^*\mathcal{G}^\bullet$$

satisfies the equivalent conditions of Lemma 0DJ8. This was shown in the proof of Lemma 0A1D. □